Arjun Mauji

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Professional Summary

Mechanical Engineering undergraduate at Queen's University with hands-on experience in mechanical and embedded systems design, along with strong problem-solving and communication skills developed through collaborative engineering projects.

Education

B.A.Sc. Mechanical Engineering, Queen's University *Kingston, Canada | 3rd/4*

- Dean's Scholar 24/25, Biggs – Ronald & Deanna Scholarship
- Relevant Coursework/Concepts: Embedded/Bare-Metal Programming, Real Time Operating Systems, Microcontrollers, Computer Architecture, Manufacturing Methods, Machine Tool Laboratory, Material Science

Work Experience

Teaching Assistant (Mechatronics Course), Queen's University Sep – Dec 2025

- Co-supervised 10 mechatronics labs with over 100 students on measurement sensors and signal noise filtering
- Answered electronic measurement theory questions and troubleshot lab circuits

Projects

Digital Oscilloscope, Individual Design Project July 2026

- Designing a digital STM32-based oscilloscope to sample and visualize signals up to 100kHz in the 0-3.3 V range
- Implementing a high impedance front end to safely sample signal voltage at 1Msps with probes
- Using DMA to store signal samples and transmit them to the display over SPI in parallel, avoiding CPU blocking
- Developing the firmware in bare-metal C, with a custom SPI driver and build process to flash code without an IDE
- Creating a custom PCB and 3D printed enclosure to house the complete electronics assembly

Embedded Software Team, Queen's Off-Road Car Design Team (Baja) Feb – Mar 2026

- Programmed STM32 boards in C to monitor the vehicle transmission system's thermal condition
- Assisted in migrating the vehicle's existing CAN bus communication library to use FreeRTOS

Hardware Motion Analyzer, Individual Design Project Nov – Dec 2025

- Prototyped a low-cost (<$40) wireless motion analyzer to compare dynamic movement flexibility between users
- Programmed 2 IMU-based sensor modules to capture movements by calculating joint flexion and extension angles
- Implemented RTOS tasking with I2C communication to log and filter up to 10 angle data streams concurrently
- Analyzed movement flexibility and peak angle values using SciPy

Mechanical Team, Queen's Off-Road Car Design Team (Baja) Sep – Dec 2024

- Assisted in the remodelling and manufacturing of the continuously variable transmission (CVT) using SolidWorks

Skills

Mechanical: SolidWorks CAD & FEA, Onshape, Reading Assembly Drawings, PID

Embedded Systems: STM32, ESP32, FreeRTOS, CAN, I2C, SPI, UART

Software Tools: C, Python, MATLAB (Simulink), Git/GitHub

Electrical Tools: Multimeters, Soldering, LTSpice, KiCad

Professional Tools: MS Outlook, Word, Excel, PowerPoint

Spoken Languages: Bilingual (Fluent) – English & French